

Quantum optical shallow networks

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Classical shallow networks are universal approximators. Given a sufficient number of neurons, they can reproduce any continuous function to arbitrary precision, with a resource cost that scales linearly in both the input size and the number of trainable parameters. In this work, we present a quantum optical protocol that implements a shallow network with an arbitrary number of neurons. Both the input data and the parameters are encoded into single-photon states. Leveraging the Hong-Ou-Mandel effect, the network output is determined by the coincidence rates measured when the photons interfere at a beam splitter, with multiple neurons prepared as a mixture of single-photon states. Remarkably, once trained, our model requires constant optical resources regardless of the number of input features and neurons.

Keywords: Quantum classifier; Quantum optical neuron; Quantum neural networks; Hong-Ou-Mandel effect;

I. INTRODUCTION

Machine learning is often framed as the task of approximating an unknown function: a continuous mapping in regression, or a decision boundary in classification (with classes at least partially separable in the feature space). Classical shallow neural networks, i.e. networks with one hidden layer of M neurons, are known to be universal function approximators [1]. In the uniform norm, they can approximate any continuous function arbitrarily well (on compact subsets of its domain) as

$$f_W(x) = \sum_{i=0}^{M-1} w_i g(W_i \cdot x + \beta_i) \text{ with } x \in \mathbb{R}^N \quad (1)$$

and $M \in \mathbb{N}$, $\{W_i \in \mathbb{R}^N : i = 0, \dots, M-1\}$ (hidden parameters), $w, \beta \in \mathbb{R}^M$ (output parameters and biases) and g a non-constant, bounded, and continuous activation function. Different formulations of this result have been proposed, generalizing it to several cases such as unbounded activation functions [2–8]. Despite several attempts to estimate M a priori, achieving the desired expressivity can be a resource-hungry task: a shallow network requires $\mathcal{O}(MN)$ computational operations. Additional costs arise in image classification, a fundamental routine in computer vision [9–12] with applications from medical diagnosis [13, 14], to security and social media analysis [15–17]. Indeed, conventional imaging protocols require photons that scale linearly in the image resolution. Research has recently focused in finding alternative implementations of neural networks, exploiting quantum mechanical effects in computational methods [18–22] or optical implementations [23–31], with applications including image classification [32–34].

In this paper, we present a quantum optical version of a classical shallow neural network. We consider the Hong-Ou-Mandel effect [35, 36], in which the probability that two photons exit a beam splitter in different modes, depends on their spectral distinguishability, with applications to kernel methods [37] and optical perceptrons [38]. In our method, the spatial modes of a pure photon encode the input features, given by diffraction on a free-space object or as data encoded by a spatial light modulator (SLM). Then, a mixture of single-photon states encodes M set of trainable parameters (the analogue of the network hidden layer). We show that the response function of this apparatus is a universal approximator, yielding

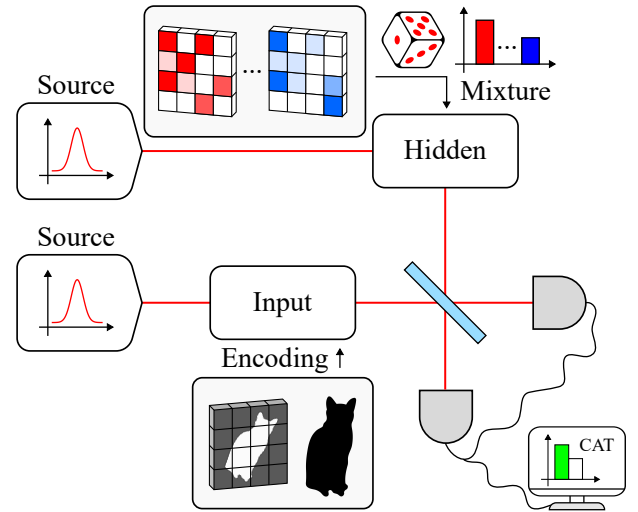


FIG. 1. Quantum optical shallow network implemented by a mixed state. One photon collects the input data, e.g. the shape of a object, the other encodes a sequence of hidden parameters, randomly sampled from a set of M different configurations with probabilities acting as output parameters. The input is classified by two bucket detectors that measure the rate of coincidences at the beam splitter output.

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Eq. (1) from the rate of photons coincidences at the output ports. See Fig. 1 for a representation of this architecture. We demonstrate that our protocol requires constant resources for a single classification instance, both in terms of photons and computational operations: a superexponential speedup over its classical counterpart. Finally, we numerically assess its approximation capabilities by verifying that increasing the number of neurons leads to improved accuracy in image classification tasks.

The spatial pattern (how the photon's amplitude varies across space) can represent an image, a pixel map, or some structured input—just like in classical optical computing, where you project an image or pattern onto light.

II. PROTOCOL

In this section we design an optical apparatus whose response function is a universal approximator according to Eq. (1). Consider two separable single-photon states, distinctively fed into the top and left input modes of a balanced beam splitter. In the left branch, labelled by mode a , a pure single-photon encodes the input state to be processed and classified, e.g. the spectral amplitude reflected off by an unknown free-space object, or a pattern encoded by a spatial light modulator. In the top branch, labelled by mode b , a single-photon density operator encodes the parameters of the hidden layer, namely M independent sets, each corresponding to a single hidden neuron of N trainable (user-controlled) parameters. We call a and b input and hidden modes, respectively. After the beam splitter, we consider two bucket detectors that click whenever a photon exits their mode, without retrieving any spatial information. Due to the Hong-Ou-Mandel effect, the coincidence probability at the output, i.e. the probability that both detector click, depends on the spectral distinguishability between the input and the hidden modes. When the hidden state is fixed, we show that measuring the rate of coincidences probability is equivalent to computing the output of the classical shallow network of Eq. (1). We report a schematic representation of this apparatus in Fig. 1

In the input branch, consider a pure multimode single-photon state, namely

$$|\mathcal{I}\rangle = \int d^2k \mathcal{I}(k) a_\omega^\dagger(k) |0\rangle. \quad (2)$$

The input $\mathcal{I}(k)$ is encoded in the photon spatial modes, with $a_\omega^\dagger(k)$ the creation operator, $|0\rangle$ the vacuum state, and k the two-dimensional momentum conjugated to the transverse coordinates on the image plane. Eq. (2) is derived by considering a monochromatic source with frequency ω . The source emits a single photon towards the encoding system, e.g. a free-space object or a SLM. Hence, $\mathcal{I}$ contains the spectral information of the source and of the input, as well as the transfer functions of the free-space propagation and of an eventual linear optical apparatus.

In the hidden branch, consider a mixed single-photon state, namely a mixture of M pure single-photon components $\{|W_{\lambda_i}\rangle : i = 0, \dots, M-1\}$ with probability

$w_i \geq 0$, each controlled by a set of N trainable parameters, such that $\{\lambda_i \in \mathbb{R}^N : i = 0, \dots, M-1\} \rightarrow |W_{\lambda_i}\rangle$. This state is described by the density operator $\rho_{\mathcal{U}_\lambda} = \sum_i w_i |W_{\lambda_i}\rangle \langle W_{\lambda_i}|$, that is

$$\rho_{\mathcal{U}_\lambda} = \int d^2k_1 d^2k_2 \mathcal{U}_\lambda(k_1, k_2) b_\omega^\dagger(k_1) |0\rangle \langle 0| b_\omega(k_2), \quad (3)$$

$$\text{with } \mathcal{U}(k_1, k_2) = \sum_{i=0}^{M-1} w_i W_{\lambda_i}(k_1) W_{\lambda_i}^*(k_2), \quad (4)$$

subject to the normalization conditions $\sum_i w_i = 1$ and $\|W_{\lambda_i}\|^2 = 1 \ \forall i$, with $\|\cdot\|$ denoting the L^2 -norm. This state can be generated directly or indirectly: by subsequently feeding the hidden branch with pure states agnostically sampled from $\rho_{\mathcal{U}_\lambda}$, with probability w_i , or by measuring the rate of coincidences of $|W_{\lambda_i}\rangle$ through M distinguishable experimental runs, taking their weighted average over w_i . Although computationally more expensive, the latter strategy provides more freedom, bypassing the normalization and positivity constraints on w_i (allowing the exploration of a larger subspace of parameters during training). At the end of the section, we discuss how different generation protocols affect the resource cost of a single classification instance.

Consider a bucket detector, namely a photodetector without spatial sensitivity, placed at the output of each branch. After feeding both states into the beam splitter, we describe coincidences in terms of the projector that finds a photon in the collective modes a and b integrated on k (bucket detectors), i.e. $\Pi_a \otimes \Pi_b$. For a single pure component $|W_{\lambda_i}\rangle$, the probability that both detector click reads $\text{Tr}[U_H \rho_i U_H^\dagger \Pi_a \otimes \Pi_b] = (1 - |\langle \mathcal{I}, W_{\lambda_i} \rangle|^2)/2$, where U_H is the beam splitter unitary and ρ_i the input-hidden bipartite density operator, corresponding to $|\mathcal{I}\rangle \otimes |W_{\lambda_i}\rangle$. For a mixture $\rho = \sum_i w_i \rho_i$, the coincidence probability reads $p(1_a \cap 1_b) = [1 - f_{wW}(\mathcal{I})]/2$, with

$$f_{wW}(\mathcal{I}) = \sum_{i=0}^{M-1} w_i |\langle \mathcal{I}, W_{\lambda_i} \rangle|^2, \quad (5)$$

and $\langle \cdot, \cdot \rangle$ the L^2 inner product. Measurements apply a nonlinear map to such product: an essential step to approximate a neuron and overcome the limitations of linear regression. See Supplementary Material A for a complete derivation from Wick's theorem. We finalize our model by post-processing the output with a bias $\beta \in \mathbb{R}$ that increases the network flexibility (for example, to counteract unbalanced training datasets), and a subsequent sigmoid activation function $\sigma(x) = 1/[1 + \exp(-x)]$, that enhances the overall nonlinearity. Namely, the post-processed output reads $F_\theta = \sigma(f_{wW}(\mathcal{I}) + \beta)$, with θ denoting the set of trainable parameters $\{w, W_\lambda, \beta\}$. Importantly, both photons have the same frequency ω , which can be always enforced by a proper choice of the longitudinal momentum. Experimental mismatches can be solved by optimizing the hardware setup, or by introducing an additional trainable parameter to improve visibility [39].

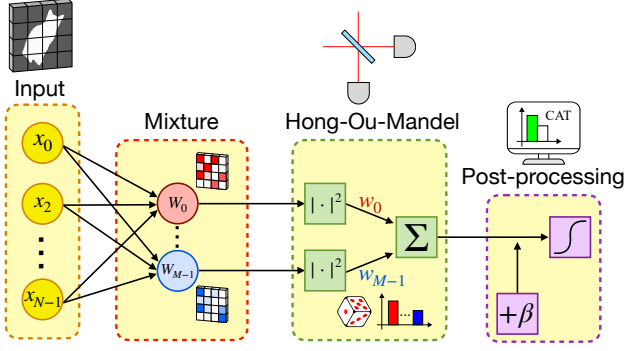


FIG. 2. Analogy between a classical shallow neural network, i.e. a network with M hidden neurons, and our interferometric setup. The nodes x_j are related to the input spectral amplitudes, encoded by the spatial light modulator on the left branch of the interferometer. Each pure component of the mixture encodes N parameters of a different hidden neuron W_i , with square absolute value activation function and no biases (a total of M sets of parameters). The output weights w_i , eventually normalized and positive, are related to the probabilities attached to each component of the mixture. Finally, a sigmoid activation function and a bias are introduced in post-processing.

Eq. (5) is precisely the response function of a classical shallow neural network. To see this, let the input and the hidden states be implemented by two amplitude-modulated SLMs of N pixels. In the left branch, N input features $\{x_j\}$ (e.g. the pixel values of a digital image) are encoded in the spatial spectrum $\hat{I}(r) = \sum_j x_j \delta(r - r_j)$, with $r \in \mathbb{R}^2$ the transverse spatial coordinate on the image plane and r_j the position of each pixel on the SLM, where $j = 0, \dots, N-1$. The hat operator denotes the two-dimensional Fourier transform on such plane. In the hidden branch, each component of $\rho_{\mathcal{U}_\lambda}$ reads $\hat{W}_{\lambda_i}(r) = \sum_j (W_{\lambda_i})_j \delta(r - r_j)$. By regularizing the Dirac delta as a sequence of approximating functions, we get $f_{wW}(\mathcal{I}) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} w_i |(W_{\lambda_i})_j x_j|^2$. Identifying $(W_{\lambda_i})_j \leftrightarrow W_{ij}$, we obtain Eq. (1) for a square absolute value activation function with no biases: a universal approximator. We report a schematic representation of this analogy in Fig. 2.

Classically, a shallow neural network (with N input features and M hidden neurons) has a computational cost that scales at least linearly in the number of mathematical operations. Indeed, a single classification instance requires the (subsequent or parallel) computation of M scalar products of vectors of size N , namely $\mathcal{O}(MN)$ resources. When features are pixels values, an additional optical cost arises due to image acquisition. In the simplest scenario, i.e. digital black-and-white images, such cost scales linearly with the resolution, namely $\mathcal{O}(N)$ photons. Our method exploits the Hong-Ou-Mandel effect, performing such resource-hungry calculations at the optical level, i.e. with constant computational resources: $\mathcal{O}(1)$ mathematical operations only determined by the

post-processing application of a single bias and a scalar sigmoid activation function [38]. Optically, the computation is not deterministic: we must take into account the number of statistical samples, i.e. photons, needed to estimate $f_{wW}(\mathcal{I})$ in terms of the probability $p(1_a \cap 1_b)$. For the i th pure state ρ_i of a mixture $\rho = \sum_i w_i \rho_i$, estimating the rate of coincidences $\text{Tr}[U_H \rho_i U_H^\dagger \Pi_a \otimes \Pi_b]$ from the empirical frequencies (coincidences over the total number of statistical samples) with uncertainty ς , requires $\mathcal{O}(\varsigma^{-2})$ photons. This cost does not depend on the number N of input features encoded in ρ_i . Consider now the mixture of M pure states. Reconstructing $p(1_a \cap 1_b)$ by separately keeping track of each pure contribution, would require $\mathcal{O}(M\varsigma^{-2})$. Surprisingly, such cost can be bypassed whenever the reconstruction is performed agnostically: without keeping track of which pure state yielded each outcome. In such case, i.e. when the mixture is generated by sampling from pure states, $p(1_a \cap 1_b)$ can be obtained as the statistical sample average of a random variable, with outcome $\text{Tr}[U_H \rho_i U_H^\dagger \Pi_a \otimes \Pi_b]$ and probability w_i for $i = 0, \dots, M-1$. From the Hoeffding's inequality [40], estimating $p(1_a \cap 1_b)$ with uncertainty ε and confidence level $1 - \delta$ requires $\mathcal{O}(\varepsilon^{-2} \log(2/\delta))$ photons. See Supplementary Material C for a rigorous derivation. Namely, the cost of a single classification instance with a quantum optical shallow network does not scale neither with the input size N nor the number of hidden neurons M .

Finally, we discuss an alternative encoding strategy for the hidden parameters. We replace the mixture of Eq. (3) with a superposition of M states $|W_{\lambda_i}\rangle$, which reads $\rho_{\mathcal{U}_\lambda} = \sum_{ij} w_i w_j |W_{\lambda_i}\rangle \langle W_{\lambda_j}|$. In this case, measuring the rate of coincidences gives $f_{wW}(\mathcal{I}) = |\sum_i w_i \langle \mathcal{I}, W_{\lambda_i} \rangle|^2$. This coherently provides a single classification instance, still with constant resources. If compared to the mixture, the square absolute value is applied after the scalar product with w_i : the model can be cast into a quadratic regression protocol by a simple rotation in the parameters space, without exhibiting universal approximation capabilities. Namely, its performances do not depend on the number of neurons in the hidden layer, making it unfeasible when learning complex patterns. See Supplementary Materials A and B for a mathematical discussion of this protocol.

III. TRAINING AND CLASSIFICATION

In this section, we discuss the training of the set of parameters $\theta = \{w, W_\lambda, \beta\}$. Then, we present simulations of our model applied to the classification of images.

We consider classification tasks in which approximators, according to Eq. (1), are trained to draw a boundary between binary classes that are (at least partially) separable in the feature space. Let D be a training set, that is, a collection of data samples $\{\mathcal{I}_s\}$ in the feature space, e.g. images represented as points whose coordinates correspond to pixel intensities, with known labels

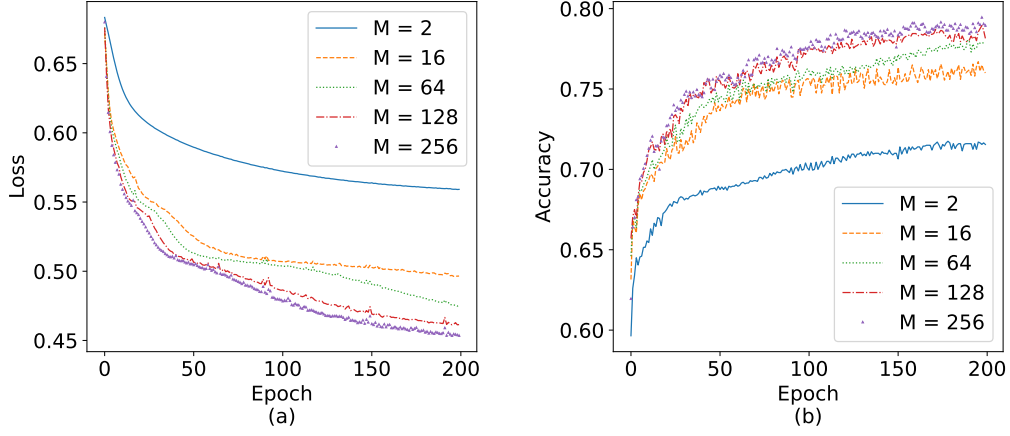


FIG. 3. History plot of the quantum optical shallow network, when classifying images of planes and dogs from the CIFAR-10 validation dataset. The optical network is modelled by an amplitude modulated hidden state of 1024 pixels. Each line is associated to a different number M of neurons of the hidden layer, i.e. a combination of multiple hidden parameters into a mixed state (a trainable convex combination of M pure state). Optimization is performed with Adam for 200 epochs, batch size 1000 and learning rate 0.03. (a) Binary cross-entropy versus the number of training epochs. (b) Accuracy versus the number of training epochs. Increasing the number of hidden neurons leads to improved asymptotic accuracy.

$\{y_s \in \{0, 1\}\}$. Each training sample is then individually and sequentially fed into the input branch of the interferometer. **Classes predicted from the rate of coincidences are compared with the target ones, using the binary-cross entropy H as loss function.** Taking into account the normalization of the hidden and the output weights, as well as the positivity of the latter, we describe training as a constrained optimization problem

$$\min_{\theta} \sum_{s \in D} H(y_s, F_{s\theta}) \quad (6)$$

$$\text{subject to } \sum_i w_i = 1, w_i \geq 0 \forall i \quad (7)$$

$$\|W_{\lambda_i}\|^2 = 1 \forall i.$$

Computing the loss function derivatives, **we optimize H via stochastic (mini-batch) gradient descent [41].** Parameters are updated according to the rule

$$\theta \rightarrow \theta - \frac{\eta}{|B|} \sum_{s \in B_l} \nabla_{\theta} H(y_s, F_{s\theta}), \quad (8)$$

with η called learning rate. **At each epoch, i.e. a full pass through the dataset, the update rule is subsequently applied to $\{B_l \subseteq D\}$, called batches, i.e. elements of a random partition of D into subsets of cardinality $|B|$.**

Gradient descent can be constrained by basically two means. On the one hand, one can enforce normalization and positivity directly in Eq. (5), modifying the response function and the gradients accordingly. This approach resembles the paradigm of weight normalization, commonly adopted in classical neural networks [42]. On the other hand, such conditions can be continuously enforced, by manually constraining the parameters after each application of Eq. (8). Although computationally less expensive, this last approach provides a limited exploration of the parameter space, dampening the model

expressivity as well as training convergence. See **Supplementary Material B** for a mathematical discussion that takes into accounts constraints. We consider a specialization of those results to the case where both the input and the hidden branches are modelled by two amplitude-modulated SLMs, enforcing the constraints after each epoch. **Applying the chain rule to Eq. (8), the only two model-dependent contributions in $\nabla_{\theta} H$ are $\nabla_w f$ and $\nabla_{\lambda_i} f$.** We obtain $(\nabla_w f)_c = \sum_{j=0}^{N-1} |(W_{\lambda_c})_j x_j|^2$ and $(\nabla_{\lambda_c} f)_d = 2w_c \sum_{j=0}^{N-1} x_d (W_{\lambda_c})_j x_j$, where $c = 0, \dots, M-1$ and $d = 0, \dots, N-1$. Finally, we reduce the number of constraints in Eq. (7), by keeping track of the output weights normalization $K = \sum_i w_i$, post-processing the output as $\tilde{F}_{\theta} = \sigma(K f_w W(\mathcal{I}) + \beta)$. Despite the increased computational training cost (but not the optical one), this strategy improves the model convergence.

For our simulations, we considered three benchmark datasets: the MNIST dataset [43], consisting of 28×28 images of handwritten digits (0-9); the Fashion-MNIST dataset [44], which contains 28×28 grayscale images of clothing items such as T-shirts, pullovers, and bags; and the CIFAR-10 dataset [45], composed of 32×32 color images spanning 10 different categories, including cats, dogs and planes. To ensure consistent input dimensions, we zero-padded the MNIST and Fashion-MNIST images to 32×32 pixels and converted CIFAR-10 images to grayscale. Each input image is directly encoded in an amplitude-modulated SLM $\hat{\mathcal{I}}(r) = \sum_j x_j \delta(r - r_j)$, with x_j the normalized pixel values. We use the **binary cross-entropy as loss function and the Adam optimizer, namely a stochastic mini-batch gradient descent accelerated with momentum [46].** We employ **Glorot initialization** for both the hidden and output weights [47]. Model performance is evaluated in terms of classification accuracy on the test set, defined as the ratio between correctly classified sam-

Accuracy (Best) [%]	MNIST	Fashion MNIST	CIFAR
$M = 2$	99.76	96.05	71.20
$M = 16$	99.90	96.70	76.25
$M = 64$	99.95	96.85	77.05
$M = 128$	99.95	96.90	78.30
$M = 256$	99.95	96.90	78.70

TABLE I. Best Classification accuracy of the quantum optical shallow network for different numbers M of hidden neurons, evaluated on the validation dataset. Training performed with Adam for 150 epochs, batch size 1000 and learning rate 0.03, on the datasets MNIST (zeros and ones), Fashion-MNIST (T-shirts/tops and pullovers) and CIFAR-10 (planes and dogs). Increasing the number of neurons improves the asymptotic accuracy (with fluctuations near the plateau).

ples and the total number of data samples. The dataset is split into training and test subsets using an 85:15 ratio.

In Fig. 3, we show the history plot when distinguishing between planes and dogs in the CIFAR-10 dataset. With $M = 2$ hidden neurons, our model reaches an asymptotic accuracy of 71% after 200 epochs. Increasing the number of hidden neurons to $M = 256$ results in a saturated accuracy of approximately 78%. This behaviour is consistent with Eq. (1), which states that increasing the complexity of the hidden layer enhances the network’s ability to approximate the decision boundary. In Table I, we report the best (i.e. maximal) test accuracy achieved across all datasets as a function of the number of hidden neurons ($M = 2, 16, 64, 128, 256$). In Supplementary Material D, we present analogous results for classical networks with comparable architecture and complexity, highlighting the reduction in accuracy caused by the normalization and positivity constraints.

IV. CONCLUSIONS

In this work, we designed a quantum optical analogue of a shallow neural network. We mathematically proved the analogy between our protocol and its classical counterpart, demonstrating its universal approximation capabilities. Increasing the number of neurons leads to improved expressivity, without modifying the number of resources. This is a superexponential speedup over classical algorithms, with comparable accuracy.

Despite supporting an arbitrary number of hidden neurons, our architecture is still constrained to scalar outputs. Increasing the output dimension would include new possible applications, from multinomial classification tasks or multivariate regressions. For example, this can be achieved by employing alternative encoding methods, multi-port optical components in place of the beam splitter, or partial photon-number-resolving detectors to

encode multiple output nodes in multiple regions of limited spatial sensitivity.

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CODE AVAILABILITY

The underlying code that generated the data for this study is openly available in GitHub [48].

The paper shows that you can build an entire shallow neural network using just two photons, a beam splitter, and the Hong–Ou–Mandel (HOM) effect. One photon encodes the input image, the other encodes all hidden layer neurons as a mixture of spatial patterns. When these photons interfere at the beam splitter, the HOM effect naturally produces a nonlinear similarity measure between the input and all weight patterns—essentially performing the dot product and activation of a neuron in a single physical step. By measuring coincidence counts as a training the weight photon’s structure classically, this system becomes a universal function approximator that performs inference at constant physical cost, independent of the number of inputs or neurons.

SUPPLEMENTARY MATERIAL A: HONG-OU-MANDEL COINCIDENCES

In this section, we compute the rate of coincidences at the output of a Hong-Ou-Mandel interferometer, when feeding a generic (single-photon) density operator ρ into one branch, and a pure state $|\mathcal{I}\rangle$ in the other. Adopting the monochromatic approximation, we assume that the wavefront propagates along definite-sign z -directions only, i.e. by considering its transversal degree of freedom only. In the input branch, we consider the single-photon state

$$|\mathcal{I}\rangle = \int d^2k \mathcal{I}_\omega(k) a_\omega^\dagger(k) |0\rangle, \quad (\text{A1})$$

where $\hat{\mathcal{I}}_\omega(k) = (\hat{\Phi}_\omega * \hat{\mathcal{O}})\hat{\mathcal{L}}_d$ is the single-photon spectrum (with source Φ) that encodes the object shape $\mathcal{O}$, after a linear optical system with transfer function $\mathcal{L}_d$, with $*$ the convolution operator. In the hidden branch, we consider the single-photon density operator

$$\rho_{\mathcal{U}_\lambda} = \int d^2k_1 d^2k_2 \mathcal{U}_\lambda(k_1, k_2) b_\omega^\dagger(k_1) |0\rangle\langle 0| b_\omega(k_2), \quad (\text{A2})$$

with λ a collection of trainable parameters. We write the input-hidden bipartite density operator as

$$|\mathcal{I}\rangle\langle \mathcal{I}| \otimes \rho_{\mathcal{U}_\lambda} = \int d^2k_1 d^2k_2 d^2k_3 d^2k_4 \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_3, k_4) a^\dagger(k_1) b^\dagger(k_3) |0\rangle\langle 0| a(k_2) b(k_4), \quad (\text{A3})$$

From now on, we omit the frequency subscript ω . The balanced beam splitter acts on each mode as

$$\begin{cases} a^\dagger \rightarrow \frac{1}{\sqrt{2}} (a^\dagger + b^\dagger) \\ b^\dagger \rightarrow \frac{1}{\sqrt{2}} (a^\dagger - b^\dagger) \end{cases}, \quad (\text{A4})$$

with output density operator

$$\rho = \frac{1}{4} \int \prod_{i=1}^4 d^2k_i \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_3, k_4) [a^\dagger(k_1) + b^\dagger(k_1)] [a^\dagger(k_3) - b^\dagger(k_3)] |0\rangle\langle 0| [a(k_2) + b(k_2)] [a(k_4) - b(k_4)]. \quad (\text{A5})$$

Detection of mode $m \in \{a, b\}$ is described by the projector $\Pi_m = \int d^2k m^\dagger(k) |0\rangle\langle 0| m(k)$. The rate of coincidences, i.e. the probability that one and only one photon is detected in each mode, reads

$$p(1_a \cap 1_b) = \text{Tr}[\rho \Pi_a \otimes \Pi_b], \quad (\text{A6})$$

$$\text{with } \Pi_a \otimes \Pi_b = \int d^2k_5 d^2k_6 a^\dagger(k_5) b^\dagger(k_6) |0\rangle\langle 0| a(k_5) b(k_6). \quad (\text{A7})$$

By substitution of Eq. (A5), we get

$$p(1_a \cap 1_b) = \frac{1}{4} \int \prod_{i=1}^6 d^2k_i \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_3, k_4) A(k), \quad (\text{A8})$$

where $k = (k_1, k_2, \dots, k_6)$ and

$$\begin{aligned} A(k) &= \langle 0| a(k_5) b(k_6) [a^\dagger(k_1) + b^\dagger(k_1)] [a^\dagger(k_3) - b^\dagger(k_3)] |0\rangle\langle 0| [a(k_2) + b(k_2)] [a(k_4) - b(k_4)] a^\dagger(k_5) b^\dagger(k_6) |0\rangle \\ &= [\delta(k_1 - k_6) \delta(k_3 - k_5) - \delta(k_1 - k_5) \delta(k_3 - k_6)] [\delta(k_2 - k_6) \delta(k_4 - k_5) - \delta(k_2 - k_5) \delta(k_4 - k_6)]. \end{aligned} \quad (\text{A9})$$

After integrating on k_5 and k_6 , we obtain

$$p(1_a \cap 1_b) = \frac{1}{2} \int \prod_{i=1}^4 d^2k_i \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_3, k_4) [\delta(k_1 - k_2) \delta(k_3 - k_4) - \delta(k_1 - k_4) \delta(k_2 - k_3)] \quad (\text{A10})$$

$$\Rightarrow p(1_a \cap 1_b) = \frac{1}{2} \left[\|\mathcal{I}\|^2 \text{Tr}[\rho_{\mathcal{U}_\lambda}] - \int d^2k_1 d^2k_2 \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_2, k_1) \right], \quad (\text{A11})$$

with $\|\cdot\|$ denoting the L^2 -norm. We define $C = \|\mathcal{I}\|^2 \text{Tr}[\rho_{\mathcal{U}_\lambda}]$, an additional hyperparameter, taking into account optical losses that can prevent normalization. We discuss Eq. (A11) in several scenarios, i.e. when $\rho_{\mathcal{U}_\lambda}$ describes a single pure state, a superposition or a mixed state.

Pure state In this section, we derive the response function of a single artificial neuron. Let $\rho_{\mathcal{U}_\lambda}$ be a (single) pure state, namely $\rho_{\mathcal{U}_\lambda} = |W_\lambda\rangle\langle W_\lambda|$ with $\mathcal{U}_\lambda(k_1, k_2) = W_\lambda(k_1)W_\lambda^*(k_2)$. By substitution in Eq. (A11), we get

$$p(1_a \cap 1_b) = \frac{1}{2} [\|\mathcal{I}\|^2 \|W_\lambda\|^2 - |\langle \mathcal{I}, W_\lambda \rangle|^2] , \quad (\text{A12})$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product in L^2 . With $\text{Tr}[\rho_{\mathcal{U}_\lambda}] = \|W_\lambda\|^2$ and $f_W(\mathcal{I}) = C - 2p(1_a \cap 1_b)$, this formula reads

$$f_W(\mathcal{I}) = |\langle \mathcal{I}, W_\lambda \rangle|^2 . \quad (\text{A13})$$

This is the response function of a single neuron (with square modulus activation function), already discussed in [38]. In this case, the neuron is provided with no bias and a square absolute value activation function, with (trainable) parameters encoded in the pure state $|W_\lambda\rangle$. Eq. (A13) requires $\mathcal{O}(1)$ computational resources and $\mathcal{O}(\varsigma^{-2})$ photons, for a single classification instance with uncertainty ς . This is a superexponential speedup over a classical artificial neuron, whose cost scales linearly in the number of input features (both optically and computationally). See [38] for a thorough analysis of the classical scenario, and **Supplementary Material C** for a discussion about optical resources.

Mixture In this section, we describe the response function of a neural network with M hidden neurons. Consider a mixed state $\rho_{\mathcal{U}_\lambda}$, namely a convex combination of pure states $\{|W_{\lambda_i}\rangle : i = 0, \dots, M-1\}$ with probabilities $\{w_i \geq 0 : i = 0, \dots, M-1\}$. Hence

$$\mathcal{U}(k_1, k_2) = \sum_{i=0}^{M-1} w_i W_{\lambda_i}(k_1) W_{\lambda_i}^*(k_2) , \quad (\text{A14})$$

$$\text{where } \sum_{i=0}^{M-1} w_i \|W_{\lambda_i}\|^2 = 1 . \quad (\text{A15})$$

By substitution in Eq. (A11), we get

$$p(1_a \cap 1_b) = \frac{1}{2} \left[\|\mathcal{I}\|^2 \text{Tr}[\rho_{\mathcal{U}_\lambda}] - \sum_{i=0}^{M-1} w_i |\langle \mathcal{I}, W_{\lambda_i} \rangle|^2 \right] , \quad (\text{A16})$$

where $\text{Tr}[\rho_{\mathcal{U}_\lambda}] = \sum_i w_i \|W_{\lambda_i}\|^2$. Assuming the normalization of $|W_{\lambda_i}\rangle$, i.e. $\|W_{\lambda_i}\|^2 = 1 \ \forall i$, Eq. (A15) simplifies to $\sum_i w_i = 1$. Writing the last equation with respect to $f_{wW}(\mathcal{I}) = C - 2p(1_a \cap 1_b)$, we obtain

$$f_{wW}(\mathcal{I}) = \sum_{i=0}^{M-1} w_i |\langle \mathcal{I}, W_{\lambda_i} \rangle|^2 . \quad (\text{A17})$$

This is precisely the output of a neural network with a single hidden layer of M neurons, i.e. a universal approximator according to Hornik-Stinchcombe-White theorem [3], with no biases and a square modulus activation function, constrained by the normalization of Eq. (A15) and the positivity of the output weights $w_i \geq 0$.

Superposition Consider $\rho_{\mathcal{U}_\lambda}$ as a superposition of pure states $\{|W_{\lambda_i}\rangle : i = 0, \dots, M-1\}$ with amplitudes $\{w_i \in \mathbb{C} : i = 0, \dots, M-1\}$. Namely

$$\mathcal{U}(k_1, k_2) = \sum_{i,j=0}^{M-1} w_i w_j^* W_{\lambda_i}(k_1) W_{\lambda_j}^*(k_2) , \quad (\text{A18})$$

$$\text{where } \text{Tr}[\rho_{\mathcal{U}_\lambda}] = \sum_{i,j=0}^{M-1} w_i w_j^* \int d^2k W_{\lambda_i}(k) W_{\lambda_j}^*(k) = 1 . \quad (\text{A19})$$

By substitution in Eq. (A11), we get

$$p(1_a \cap 1_b) = \frac{1}{2} \left[\|\mathcal{I}\|^2 \text{Tr}[\rho_{\mathcal{U}_\lambda}] - \sum_{i,j=0}^{M-1} w_i w_j^* \langle \mathcal{I}, W_{\lambda_i} \rangle \langle W_{\lambda_j}, \mathcal{I} \rangle \right] , \quad (\text{A20})$$

$$f_{wW}(\mathcal{I}) = \left| \sum_{i=0}^{M-1} w_i \langle \mathcal{I}, W_{\lambda_i} \rangle \right|^2 , \quad (\text{A21})$$

with $f_{wW}(\mathcal{I}) = C - 2p(1_a \cap 1_b)$. From now on, we assume that $w_i \in \mathbb{R}$. This model represents an alternative encoding scheme for the hidden parameters, namely a coherent implementation that has the same resource costs of the mixture. Several issues arise in this scenario. Mathematically, Eq. (A21) is not a universal approximator, in the sense of Hornik-Stinchcombe-White theorem [3], since it describes M hidden neurons but with linear activation functions. A single instance of Eq. (A21) can be reparametrized, i.e. collapsed, to that of a single neuron. In quantum mechanical terms: superposition can be viewed as a change of basis on the experimental apparatus implemented by a single pure component $|W_{\lambda_i}\rangle$. This implies limited learning capabilities, independently on the network complexity M . Such limitations can also be confirmed numerically. In our simulations, this model did not exhibit the same accuracy scalability of the mixture (with respect to the number M of hidden neurons).

SUPPLEMENTARY MATERIAL B: LOSS FUNCTION DERIVATIVES

In this section, we discuss training for the quantum optical shallow network. In particular, we compute the derivatives associated to the implementations discussed in the previous section. First, we describe gradients in general terms. Then, we specialize to Eqs. (A17) and (A21). Let us define the set of training parameters $\theta = \{w, W_\lambda, \beta\}$, where $w \in \mathbb{R}^M$ are the output weights, $\beta \in \mathbb{R}$ the output bias and $\{W_{\lambda_i} \in \mathbb{R}^N : i = 0, \dots, M-1\}$ the weights of the hidden layer, i.e. a set of M neurons each with N weights such that $W_\lambda \in \mathbb{R}^{M \times N}$. Training is performed by applying an Adam-optimized mini-batch gradient descent on the binary cross-entropy

$$H(y_s, F_{s\theta}) = -y_s \log F_{s\theta} - (1 - y_s) \log (1 - F_{s\theta}), \quad (\text{B1})$$

where $y_s \in \{0, 1\}$ is the ideal output (i.e. correct class label) of the s th data sample of the training dataset D , while $F_{s\theta} \in (0, 1)$ is the corresponding predicted label, computed as

$$F_{s\theta} = g(f_{wW}(\mathcal{I}_s) + \beta), \quad (\text{B2})$$

with g the activation function, e.g. the sigmoid $\sigma(x) = 1/[1 + \exp(-x)]$, and $\mathcal{I}_s$ the s th sample of the dataset. Here, f_{wW} represents the output of the models, given by Eq. (A17) or Eq. (A21), and obtained from the probabilities of the Hong-Ou-Mandel coincidences. Independently of the model, Eq. (A11) reads

$$f_{wW}(\mathcal{I}) = \int d^2 k_1 d^2 k_2 \mathcal{I}(k_1) \mathcal{I}^*(k_2) \mathcal{U}_\lambda(k_2, k_1). \quad (\text{B3})$$

We compute the derivative of the loss function by applying the chain rule $\nabla_\theta H = \partial_F H \partial_\xi g \nabla_\theta \xi$, where $\xi = f_{wW} + \beta$. For each set of parameters we get $\partial_\beta \xi = 1$ and

$$\begin{aligned} \nabla_w \xi &= \nabla_w f = \int d^2 k_1 d^2 k_2 \mathcal{I}(k_1) \mathcal{I}^*(k_2) \nabla_w \mathcal{U}(k_2, k_1), \\ \nabla_{\lambda_c} \xi &= \nabla_{\lambda_c} f = \int d^2 k_1 d^2 k_2 \mathcal{I}(k_1) \mathcal{I}^*(k_2) \nabla_{\lambda_c} \mathcal{U}(k_2, k_1), \end{aligned} \quad (\text{B4})$$

where $\nabla_{\lambda_c} \in \{\nabla_{\lambda_0}, \dots, \nabla_{\lambda_{M-1}}\}$ labels the derivative with respect to the c th hidden neuron, each with N components labelled by d , i.e. $(\nabla_{\lambda_c} f)_d \in \{(\nabla_{\lambda_c} f)_0, \dots, (\nabla_{\lambda_c} f)_{N-1}\}$. Similarly, $(\nabla_w f)_c \in \{(\nabla_w f)_0, \dots, (\nabla_w f)_{M-1}\}$. From now on, we guarantee differentiability by assuming that all the parameters are real. Otherwise, the binary-cross entropy (which is real-valued) would not be holomorphic. After computing the loss function derivative, the parameters are updated according to the rule

$$\theta \rightarrow \theta - \frac{\eta}{|B|} \sum_{s \in B_l} \nabla_\theta H(y_s, F_{s\theta}), \quad (\text{B5})$$

with η called learning rate. At each epoch, the update rule is subsequently applied for each batch $B_l \subseteq D$, obtained by randomly dividing the dataset in partitions of $|B|$ data samples, with cardinality $|B|$.

Mixture In this section, we explicitly discuss the mixture case. After each epoch, the parameters are updated and normalization needs to be guaranteed by Eq. (A15). We do this, by enforcing such condition directly in f_{wW} , modifying the derivative while keeping track of the additional terms. Classically, this resembles the concept of weight normalization [42]. Hence, Eq. (A17) reads

$$\mathcal{U}(k_1, k_2) = \sum_{i=0}^{M-1} \frac{P(w_i)}{\sum_j P(w_j)} \frac{W_{\lambda_i}(k_1) W_{\lambda_i}^*(k_2)}{\|W_{\lambda_i}\|^2}, \quad (\text{B6})$$

where P is a function enforcing the positivity of the output weights, i.e. $P(w_i) \geq 0 \forall i$. For example $P \in \{|\cdot|, \sigma, \text{ReLU}\}$. Taking the derivative with respect to the c th output weight, we get

$$(\nabla_w \mathcal{U})_c = - \sum_{p=0}^{M-1} \frac{\partial_{w_c} P(w_p)}{\sum_j P(w_j)} \sum_{i=0}^{M-1} \frac{P(w_i)}{\sum_j P(w_j)} \frac{W_{\lambda_i}(k_1) W_{\lambda_i}^*(k_2)}{\|W_{\lambda_i}\|^2} + \sum_{i=0}^{M-1} \frac{\partial_{w_c} P(w_i)}{\sum_j P(w_j)} \frac{W_{\lambda_i}(k_1) W_{\lambda_i}^*(k_2)}{\|W_{\lambda_i}\|^2} \quad (\text{B7})$$

$$= \frac{P'(w_c)}{\sum_j P(w_j)} \left[-\mathcal{U}(k_1, k_2) + \frac{W_{\lambda_c}(k_1) W_{\lambda_c}^*(k_2)}{\|W_{\lambda_c}\|^2} \right]. \quad (\text{B8})$$

In the second step, we used that $\partial_{w_c} P(w_i) = \delta_{ci} P'(w_i)$, with δ_{ci} denoting the Kronecker delta and $P'(w_i) = \partial_{w_i} P(w_i)$. Hence, $\sum_p \partial_{w_c} P(w_p) = P'(w_c)$. By substituting in Eq. (B4), we obtain

$$(\nabla_w f)_c = \frac{P'(w_c)}{\sum_i P(w_j)} \left[-f_{wW}(\mathcal{I}) + \left| \left\langle \mathcal{I}, \frac{W_{\lambda_c}}{\|W_{\lambda_c}\|} \right\rangle \right|^2 \right]. \quad (\text{B9})$$

Computing the derivative with respect to the hidden parameters, we get

$$(\nabla_{\lambda_c} \mathcal{U})_d = \frac{P(w_c)}{\sum_j P(w_j) \|W_{\lambda_c}\|^2} \left[-\frac{2W_{\lambda_c}(k_1) W_{\lambda_c}^*(k_2)}{\|W_{\lambda_c}\|^2} \text{Re}[\langle W_{\lambda_c}, (\nabla_{\lambda_c} W_{\lambda_c})_d \rangle] + [\nabla_{\lambda_c} (W_{\lambda_c}(k_1) W_{\lambda_c}^*(k_2))]_d \right]. \quad (\text{B10})$$

By substitution in Eq. (B4), we finally obtain

$$(\nabla_{\lambda_c} f)_d = \frac{2P(w_c)}{\sum_j P(w_j)} \text{Re} \left[\left\langle \mathcal{I}, \frac{(\nabla_{\lambda_c} W_{\lambda_c})_d}{\|W_{\lambda_c}\|} \right\rangle \left\langle \frac{W_{\lambda_c}}{\|W_{\lambda_c}\|}, \mathcal{I} \right\rangle - \left| \left\langle \mathcal{I}, \frac{W_{\lambda_c}}{\|W_{\lambda_c}\|} \right\rangle \right|^2 \left\langle \frac{W_{\lambda_c}}{\|W_{\lambda_c}\|}, \frac{(\nabla_{\lambda_c} W_{\lambda_c})_d}{\|W_{\lambda_c}\|} \right\rangle \right]. \quad (\text{B11})$$

As an explicit example, we consider the (amplitude- or phase-modulated) encoding provided by a spatial light modulator (SLM) of N pixels. The mixture can be generated by considering M independent patterns on the SLM, and randomly sampled with probabilities $\{w_i\}$. We denote the set of amplitudes and phases as $\{\alpha_{ij}\}$ and $\{\varphi_{ij}\}$, respectively, where $i = 0, \dots, M-1$ labels the pure components of the mixture and $j = 0, \dots, N-1$ each pixel position. Namely

$$(\lambda_c)_d = (\alpha_{cd}, \varphi_{cd}) \rightarrow \hat{W}_{\lambda_c}(r) = \sum_{d=0}^{N-1} \alpha_{cd} \exp(i\varphi_{cd}) \delta(r - r_d), \quad (\text{B12})$$

where $r \in \mathbb{R}^2$ labels the transverse coordinate on the SLM plane. We consider the same representation for the input branch, in which a digital image $\{\mathcal{I}_d\}$ is encoded as $\hat{\mathcal{I}}(r) = \sum_d \mathcal{I}_d \delta(r - r_d)$. Hence, the terms in Eqs. (B9) and (B10) simplify as

$$\langle W_{\lambda_c}, (\nabla_{\alpha_c} W_{\lambda_c})_d \rangle = \alpha_{cd}, \quad \langle W_{\lambda_c}, (\nabla_{\varphi_c} W_{\lambda_c})_d \rangle = i\alpha_{cd}^2, \quad (\text{B13})$$

$$\langle \mathcal{I}, (\nabla_{\alpha_c} W_{\lambda_c})_d \rangle = \mathcal{I}_d^* \exp(i\varphi_{cd}), \quad \langle \mathcal{I}, (\nabla_{\varphi_c} W_{\lambda_c})_d \rangle = i\mathcal{I}_d^* \alpha_{cd} \exp(i\varphi_{cd}). \quad (\text{B14})$$

We can set $\varphi_{ij} = 0$ for amplitude encoding, or $\alpha_{ij} = 1/N$ for phase encoding. For example, in the former case we obtain

$$(\nabla_{\lambda_c} f)_d = \frac{2P(w_c)}{\sum_j P(w_j) \|W_{\lambda_c}\|^2} \text{Re} \left[\mathcal{I}_d^* \langle W_{\lambda_c}, \mathcal{I} \rangle - \alpha_{cd} \left| \left\langle \mathcal{I}, \frac{W_{\lambda_c}}{\|W_{\lambda_c}\|} \right\rangle \right|^2 \right]. \quad (\text{B15})$$

Superposition We perform a similar analysis on Eq. (A21). We enforce the normalization condition directly in f_{wW} , modifying the derivative while keeping track of the additional terms. Moreover, we assume that the output weights are real, i.e. $w_i \in \mathbb{R}$, guaranteeing the differentiability of the loss function. Hence, the hidden state reads

$$\mathcal{U}(k_1, k_2) = \frac{\sum_{ij=0}^{M-1} w_i w_j W_{\lambda_i}(k_1) W_{\lambda_j}^*(k_2)}{\sum_{pq=0}^{M-1} w_p w_q \int d^2k W_{\lambda_p}(k) W_{\lambda_q}^*(k)}, \quad (\text{B16})$$

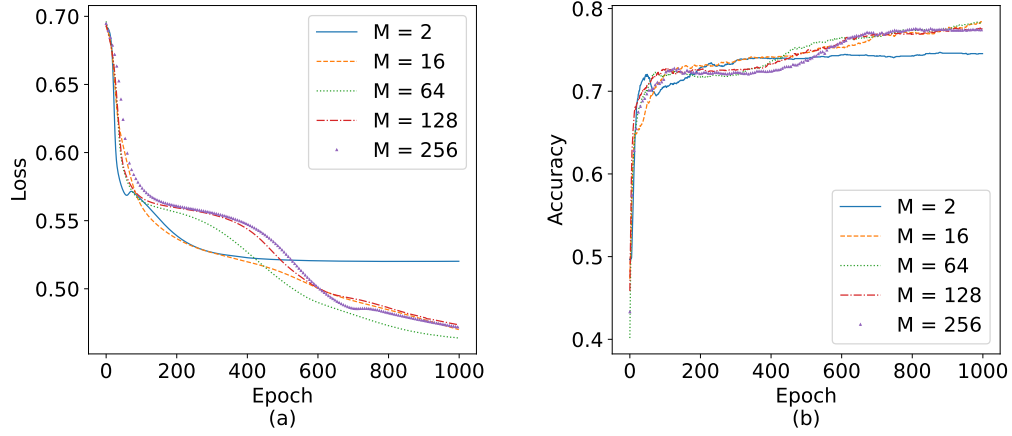


FIG. S1. History plot of an unconstrained classical shallow network, when classifying images of planes and dogs from the CIFAR-10 validation dataset. Each line is associated to a different number M of hidden neurons. Optimization is performed for 1000 epochs, batch size 1000 and learning rate 0.03. (a) Binary cross-entropy versus the number of training epochs. (b) Accuracy versus the number of training epochs.

where we employed the normalization condition of Eq. (A19). As for the mixture, we compute the derivatives of $\mathcal{U}$ and f with respect to the output weights w , yielding

$$\begin{aligned}
 (\nabla_w \mathcal{U})_c &= \sum_{i=0}^{M-1} \frac{w_i}{\text{Tr}[\rho_{\mathcal{U}_\lambda}]} \left[-2 \int d^2 k \text{Re}[W_{\lambda_c}(k) W_{\lambda_i}^*(k)] \mathcal{U}(k_1, k_2) + W_{\lambda_c}(k_1) W_{\lambda_i}^*(k_2) + W_{\lambda_i}(k_1) W_{\lambda_c}^*(k_2) \right], \\
 \Rightarrow (\nabla_w f)_c &= \sum_{i=0}^{M-1} \frac{2w_i}{\text{Tr}[\rho_{\mathcal{U}_\lambda}]} \text{Re}[-\langle W_{\lambda_i}, W_{\lambda_c} \rangle f_{wW} + \langle \mathcal{I}, W_{\lambda_c} \rangle \langle W_{\lambda_i}, \mathcal{I} \rangle].
 \end{aligned} \tag{B17}$$

Similarly, the derivative with respect to the set of hidden weights reads

$$\begin{aligned}
 (\nabla_{\lambda_c} \mathcal{U})_d &= \sum_{i=0}^{M-1} \frac{w_i}{\text{Tr}[\rho_{\mathcal{U}_\lambda}]} \left[-2w_c \int d^2 k \text{Re}[(\nabla_{\lambda_c} W_{\lambda_c})_d(k) W_{\lambda_i}^*(k)] \mathcal{U}(k_1, k_2) + \sum_j w_j \left[\nabla_{\lambda_c} (W_{\lambda_i}(k_1) W_{\lambda_j}^*(k_2)) \right]_d \right], \\
 \Rightarrow (\nabla_{\lambda_c} f)_d &= \sum_{i=0}^{M-1} \frac{2w_i w_c}{\text{Tr}[\rho_{\mathcal{U}_\lambda}]} \text{Re}[-\langle W_{\lambda_i}, (\nabla_{\lambda_c} W_{\lambda_c})_d \rangle f_{wW} + \langle \mathcal{I}, (\nabla_{\lambda_c} W_{\lambda_c})_d \rangle \langle W_{\lambda_i}, \mathcal{I} \rangle].
 \end{aligned} \tag{B18}$$

We can simplify this equation, by employing the SLM model discussed in the previous section. For amplitude encoding, i.e. $\varphi_{ij} = 0$, Eqs. (B13) and (B14) imply

$$(\nabla_{\lambda_c} f)_d = \sum_{i=0}^{M-1} \frac{2w_i w_c}{\text{Tr}[\rho_{\mathcal{U}_\lambda}]} \text{Re}[\mathcal{I}_d^* \langle W_{\lambda_i}, \mathcal{I} \rangle - \alpha_{cd} f_{wW}]. \tag{B19}$$

SUPPLEMENTARY MATERIAL C: STATISTICAL SAMPLE COMPLEXITY AND RESOURCES

In principle, our protocol requires only two photons for a single classification instance, namely $\mathcal{O}(1)$ constant resources. However, predicted classes are obtained from the rate of coincidences, i.e. a probability to be estimated by gathering multiple statistical samples, which correspond to different experimental repetitions. In this section, we discuss the number of statistical samples, i.e. photons, needed to estimate $f_{wW}(\mathcal{I})$, i.e. the output of the quantum optical shallow network with a mixed hidden state. We first review the cost of estimating the rate of coincidences for a single pure hidden state $|W_{\lambda_i}\rangle$. Then, we generalize our results when taking into account the sampling cost for a mixture of M pure states.

In our estimations, we leverage the concept of sample complexity [49, 50]. Namely, we estimate the number of statistical samples n needed to predict the target label with a given uncertainty. Consider the i th pure component ρ_i of the mixed state before the measurement. For an input state with N features (e.g. a digital image of N

pixels), the coincidence measurement corresponds to the outcome of a single optical neuron with N parameters, i.e. $|\langle \mathcal{I}, W_{\lambda_i} \rangle|^2 = 1 - 2p_i$, with $p_i \in [0, 1/2]$. Let n_i be the number of photons needed to estimate p_i . The coincidence probability follows a binomial distribution (with success-failure outcomes corresponding to the presence-absence of a coincidence event). Using the empirical frequencies f_i , the probability can be estimated as

$$p_i = f_i \pm \varepsilon_i \text{ with } \varepsilon_i = 2\sqrt{\frac{f_i(1-f_i)}{n_i}}, \quad (\text{C1})$$

where the factor 2 approximately guarantees a 0.95 confidence level. Since $f_i(1-f_i) \leq 1/4$, inverting this relation yields $n_i \geq \varepsilon_i^{-2}$. Requiring $\varsigma = \varepsilon_i \forall i$, the sample complexity of each pure state is $\mathcal{O}(\varsigma^{-2})$. Such bound does not depend on the number of features (e.g. pixels) encoded in ρ_i .

We now derive the total cost of estimating the rate of coincidences $p(1_a \cap 1_b)$ of the mixed state $\rho = \sum_i w_i \rho_i$, with probabilities $w_i \geq 0$. Intuitively, a reconstruction of $p(1_a \cap 1_b)$ that separately keeps track of each pure contribution would cost $\mathcal{O}(M\varsigma^{-2})$, namely M independent experimental repetitions (each time with a different ρ_i). This is the same cost of reconstructing a multinomial probability distribution, which scales linearly with the number of possible outcomes (bins), i.e. $\mathcal{O}(M)$. Surprisingly, this is not the case since we are interested only in estimating the statistical sample average of the rate coincidences sampled on ρ (without tracking which pure state yielded each outcome). Indeed, $f_{wW}(\mathcal{I}) = 1 - 2p(1_a \cap 1_b)$, with

$$p(1_a \cap 1_b) = \sum_{i=0}^{M-1} w_i p_i \text{ with } \sum_i w_i = 1 \text{ and } w_i \geq 0 \forall i. \quad (\text{C2})$$

Namely, the output of the optical shallow network can be estimated from the statistical sample average of a random variable, bounded in $[0, 1/2]$, with outcomes $p_0, \dots, p_{M-1}$ and probabilities $w_0, \dots, w_{M-1}$. We can estimate the sample complexity using the Hoeffding's inequality (also related to the Chernoff bound) [40].

Theorem 1 (Hoeffding's inequality). Let $X_0, \dots, X_{n-1}$ be n independent random variables (with finite first and second moments) bounded as $X_i \in [0, 1] \forall i$. For the statistical sample average $\bar{X} = \sum_i X_i/n$ with expectation value $\mathbb{E}[\bar{X}]$ and for uncertainty ε with $0 < \varepsilon < 1 - \mathbb{E}[\bar{X}]$, we get

$$\Pr(\bar{X} - \mathbb{E}[\bar{X}] \geq \varepsilon) = \Pr(\sum_i X_i - n\mathbb{E}[\bar{X}] \geq n\varepsilon) \leq \exp(-2n\varepsilon^2). \quad (\text{C3})$$

In our case, the random variables are identically distributed, yielding $\mathbb{E}[\bar{X}] = \mathbb{E}[X] = p(1_a \cap 1_b)$. For the confidence level $1 - \delta$, with $\delta = \Pr(|\bar{X} - p(1_a \cap 1_b)| \geq \varepsilon)$, we get

$$\delta = \Pr(\bar{X} - p(1_a \cap 1_b) \geq \varepsilon) + \Pr(-\bar{X} + p(1_a \cap 1_b) \geq \varepsilon) \quad (\text{C4})$$

$$= \Pr(\bar{X} - p(1_a \cap 1_b) \geq \varepsilon) + \Pr(\bar{X} - p(1_a \cap 1_b) \leq -\varepsilon) \leq 2\exp(-2n\varepsilon^2). \quad (\text{C5})$$

Once inverted, this expression gives $n \leq \varepsilon^{-2} \log(2/\delta)/2$. Hence, estimating $f_{wW}(\mathcal{I})$ from the rate of coincidences with uncertainty ε and confidence level $1 - \delta$, asymptotically requires $\mathcal{O}(\varepsilon^{-2} \log(2/\delta))$ statistical samples. Namely, resources (photons) that do not scale neither with the input size N nor the number of hidden neurons M . For example, consider a bounded random variable X with M possible outcomes, each taking values in the interval $[0, 1/2]$. As a worst-case scenario, let X be uniformly distributed. With a bootstrap one can observe that, for a given n , the variance of the distribution of the sample average $\bar{X}$ is constant with respect to M . Namely, that the number of resources needed to estimate $\mathbb{E}(X)$ depends only on the desired accuracy and confidence level, not on the number M of outcomes.

SUPPLEMENTARY MATERIAL D: CLASSICAL NEURAL NETWORK COUNTERPART

We discuss the classical counterpart of our quantum optical network, in two different scenarios. Our model imposes strict normalization constraints: each pure state must be normalized, while the mixture probabilities must be positive and sum to one. This corresponds to a L^2 normalization on the hidden parameters, and a positivity and L^1 constraints on the output weights, respectively. We benchmark the classical model both with and without such constraints.

In our simulations, we improved the classical model performances by fine-tuning the sigmoid activation function as $\tilde{\sigma}(x) = 1/[1 + \exp(-11x + 5.5)]$. Fig. S1 shows the results obtained for the classical model without constraints. The classification task remains the same as in Fig. 3: distinguishing between dog and plane images from the CIFAR-10 dataset. The results closely resemble those of the quantum optical network, with comparable accuracy and scaling with respect to number of hidden neurons. In Table S1a, we report the performance of the unconstrained classical

model, while S1b shows the results in the constrained scenario. We note that the classical network can be successfully trained in both scenarios, although with different asymptotic accuracy.

Accuracy (Best) [%]	MNIST	Fashion MNIST	CIFAR
M = 2	99.95	96.90	74.70
M = 16	99.95	96.80	78.45
M = 64	99.91	96.75	78.35
M = 128	99.95	96.80	77.65
M = 256	99.91	96.70	77.60

(a)

Accuracy (Best) [%]	MNIST	Fashion MNIST	CIFAR
M = 2	99.86	96.45	71.00
M = 16	99.86	96.55	74.50
M = 64	99.86	96.50	74.85
M = 128	99.91	96.50	75.05
M = 256	99.91	96.55	74.65

(b)

TABLE S1. Best classification accuracy of a classical shallow network, for different numbers M of hidden neurons, evaluated on the validation dataset. Training is performed with mini-batch gradient descent (without Adam) for 1000 epochs, batch size 1000 and learning rate 0.03, on the datasets MNIST (zeros and ones), Fashion-MNIST (T-shirts/tops and pullovers) and CIFAR-10 (planes and dogs). (a) Unconstrained model. (b) Constrained model. Data simulated with TensorFlow [51].

REFERENCES

- [1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning* (MIT Press, 2016) <http://www.deeplearningbook.org>.
- [2] G. Cybenko, Approximation by superpositions of a sigmoidal function, *Math. Control Signals Syst.* **2**, 303 (1989).
- [3] K. Hornik, M. Stinchcombe, and H. White, Multilayer feedforward networks are universal approximators, *Neural Netw.* **2**, 359 (1989).
- [4] K. Hornik, Approximation capabilities of multilayer feedforward networks, *Neural Netw.* **4**, 251 (1991).
- [5] H. N. Mhaskar and C. A. Micchelli, Approximation by superposition of sigmoidal and radial basis functions, *Adv. Appl. Math.* **13**, 350 (1992).
- [6] M. Leshno, V. Y. Lin, A. Pinkus, and S. Schocken, Multilayer feedforward networks with a nonpolynomial activation function can approximate any function, *Neural Netw.* **6**, 861 (1993).
- [7] A. R. Barron, Universal approximation bounds for superpositions of a sigmoidal function, *IEEE Trans. Inf. Theory* **39**, 930 (1993).
- [8] T. Chen and H. Chen, Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems, *IEEE Trans. Neural Netw.* **6**, 911 (1995).
- [9] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, Gradient-based learning applied to document recognition, *Proc. IEEE* **86**, 2278 (1998).
- [10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, ImageNet classification with deep convolutional neural networks, *Commun. ACM* **60**, 84–90 (2017).
- [11] K. He, X. Zhang, S. Ren, and J. Sun, Deep residual learning for image recognition, in *IEEE Conference on Computer Vision and Pattern Recognition*, CVPR '16, p. 770.
- [12] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, in *International Conference on Learning Representations*, ICLR '21, [arXiv:2010.11929](https://arxiv.org/abs/2010.11929).
- [13] L. Cai, J. Gao, and D. Zhao, A review of the application of deep learning in medical image classification and segmentation, *Ann. Transl. Med.* **8**, 713 (2020).
- [14] L. Dao and N. Q. Ly, Recent advances in medical image classification, *Int. J. Adv. Comput. Sci.* **15**, 7 (2024).
- [15] R. G. Babu, K. Dhineshkumar, R. Sharma, and R. Krishnamoorthy, A survey of machine learning techniques using for image classification in home security, in *IOP Conference Series: Materials Science and Engineering*, Vol. 1055 (2021) p. 012088.
- [16] Z. Long, R. McCreddie, and M. Imran, Crisisvit: A robust vision transformer for crisis image classification (2024), [arXiv:2401.02838](https://arxiv.org/abs/2401.02838).
- [17] I. Amerini, C.-T. Li, and R. Caldelli, Social network identification through image classification with CNN, *IEEE Access* **7**, 35264 (2019).
- [18] S. Lloyd, M. Mohseni, and P. Rebentrost, Quantum algorithms for supervised and unsupervised machine learning (2013), [arXiv:1307.0411](https://arxiv.org/abs/1307.0411) [quant-ph].
- [19] X.-D. Cai, D. Wu, Z.-E. Su, M.-C. Chen, X.-L. Wang, L. Li, N.-L. Liu, C.-Y. Lu, and J.-W. Pan, Entanglement-based machine learning on a quantum computer, *Phys. Rev. Lett.* **114**, 110504 (2015).
- [20] F. Tacchino, C. Macchiavello, D. Gerace, and D. Bajoni, An artificial neuron implemented on an actual quantum processor, *Npj Quantum Inf.* **5**, 26 (2019).
- [21] F. Benatti, S. Mancini, and S. Mangini, Continuous variable quantum perceptron, *Int. J. Quantum Inf.* **17**, 1941009 (2019).
- [22] S. Mangini, F. Tacchino, D. Gerace, C. Macchiavello, and D. Bajoni, Quantum computing model of an artificial neuron with continuously valued input data, *Mach. Learn.: Sci. Technol.* **1**, 045008 (2020).
- [23] G. R. Steinbrecher, J. P. Olson, D. Englund, and J. Carolan, Quantum optical neural networks, *Npj Quantum*

- [Inf.](#) **5**, 60 (2019).
- [24] N. Killoran, T. R. Bromley, J. M. Arrazola, M. Schuld, N. Quesada, and S. Lloyd, Continuous-variable quantum neural networks, [Phys. Rev. Res.](#) **1**, 033063 (2019).
 - [25] X. Sui, Q. Wu, J. Liu, Q. Chen, and G. Gu, A review of optical neural networks, [IEEE Access](#) **8**, 70773 (2020).
 - [26] A. Zhang, H. Zhan, J. Liao, K. Zheng, T. Jiang, M. Mi, P. Yao, and L. Zhang, Quantum verification of NP problems with single photons and linear optics, [Light Sci. Appl.](#) **10**, 169 (2021).
 - [27] J. Spall, X. Guo, and A. I. Lvovsky, Hybrid training of optical neural networks, [Optica](#) **9**, 803 (2022).
 - [28] D. Stanev, N. Spagnolo, and F. Sciarrino, Deterministic optimal quantum cloning via a quantum-optical neural network, [Phys. Rev. Res.](#) **5**, 013139 (2023).
 - [29] C. Wood, S. Shrapnel, and G. J. Milburn, A Kerr kernel quantum learning machine (2024), [arXiv:2404.01787 \[quant-ph\]](#).
 - [30] J. Spall, X. Guo, and A. I. Lvovsky, Training neural networks with end-to-end optical backpropagation, [Adv. Photonics](#) **7**, 016004 (2025).
 - [31] F. Hoch, E. Caruccio, G. Rodari, T. Francalanci, A. Suprano, T. Giordani, G. Carvacho, N. Spagnolo, S. Koudia, M. Proietti, *et al.*, Quantum machine learning with adaptive boson sampling via post-selection, [Nat. Commun.](#) **16**, 902 (2025).
 - [32] D. Slabbert and F. Petruccione, Classical-quantum approach to image classification: Autoencoders and quantum SVMs, [AVS Quantum Sci.](#) **7**, 023804 (2025).
 - [33] Y. Sun, D. Li, Q. Xiang, Y. Yuan, Z. Hu, X. Hua, Y. Jiang, Y. Zhu, and Y. Fu, Scalable quantum convolutional neural network for image classification, [Phys. A: Stat. Mech. Appl.](#) **657**, 130226 (2025).
 - [34] A. Sakurai, A. Hayashi, W. J. Munro, and K. Nemoto, Quantum optical reservoir computing powered by boson sampling, [Opt. Quantum](#) **3**, 238 (2025).
 - [35] C.-K. Hong, Z.-Y. Ou, and L. Mandel, Measurement of subpicosecond time intervals between two photons by interference, [Phys. Rev. Lett.](#) **59**, 2044 (1987).
 - [36] A. M. Brańczyk, Hong-Ou-Mandel interference (2017), [arXiv:1711.00080](#).
 - [37] C. Bowie, S. Shrapnel, and M. J. Kewming, Quantum kernel evaluation via Hong–Ou–Mandel interference, [Quantum Sci. Technol.](#) **9**, 015001 (2023).
 - [38] S. Roncallo, A. R. Morgillo, C. Macchiavello, L. Maccone, and S. Lloyd, Quantum optical classifier with superexponential speedup, [Commun. Phys](#) **8**, 147 (2025).
 - [39] N. Crum, M. M. Hassan, A. Green, and G. Siopsis, Mode distinguishability in multi-photon interference (2025), [arXiv:2501.14915](#).
 - [40] W. Hoeffding, Probability inequalities for sums of bounded random variables, [J. Am. Stat. Assoc.](#) **58**, 13 (1963).
 - [41] H. Robbins and S. Monro, A stochastic approximation method, [Ann. Math. Stat.](#) **22**, 400 (1951).
 - [42] T. Salimans and D. P. Kingma, Weight normalization: A simple reparameterization to accelerate training of deep neural networks, in [Adv. Neural Inf. Process.](#), Vol. 29 (2016).
 - [43] Y. LeCun, C. Cortes, and C. J. Burges, <https://www.tensorflow.org/datasets/catalog/mnist>.
 - [44] H. Xiao, K. Rasul, and R. Vollgraf, https://www.tensorflow.org/datasets/catalog/fashion_mnist.
 - [45] A. Krizhevsky, <https://www.tensorflow.org/datasets/catalog/cifar10>.
 - [46] D. P. Kingma and J. Ba, Adam: A method for stochastic optimization (2014), [arXiv:1412.6980](#).
 - [47] X. Glorot and Y. Bengio, Understanding the difficulty of training deep feedforward neural networks, in [Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics](#), Vol. 9 (2010) pp. 249–256.
 - [48] A. R. Morgillo and S. Roncallo, <https://github.com/simoneroncallo/quantum-optical-network>.
 - [49] S. Arunachalam and R. De Wolf, Optimal quantum sample complexity of learning algorithms, [J. Mach. Learn. Res.](#) **19**, 1 (2018).
 - [50] H.-C. Cheng, N. Datta, N. Liu, T. Nuradha, R. Salzmann, and M. M. Wilde, An invitation to the sample complexity of quantum hypothesis testing, [npj Quantum Inf.](#) **11**, 1 (2025).
 - [51] M. Abadi et al., [TensorFlow: Large-scale machine learning on heterogeneous systems](#) (2015).